



Experiment 15: Merge Sort

Aim

To write a program to **sort elements using Merge Sort technique**.

Theory

Merge Sort is a **divide and conquer algorithm**:

- Divide the array into two halves
- Recursively sort both halves
- Merge the sorted halves

 Time Complexity: **$O(n \log n)$**

Algorithm

MergeSort(arr, low, high)

1. if (low < high)
2. mid = (low + high) / 2
3. MergeSort(arr, low, mid)
4. MergeSort(arr, mid+1, high)
5. Merge(arr, low, mid, high)

Merge(arr, low, mid, high)

1. Create two arrays L and R
 2. Copy left half into L
 3. Copy right half into R
 4. Compare elements of L and R
 5. Insert smaller element into original array
 6. Repeat until all elements are merged
-



Simple C Program

```
#include <stdio.h>
void merge(int arr[], int low, int mid, int high) {
    int i, j, k;
    int n1 = mid - low + 1;
    int n2 = high - mid;
    int L[50], R[50];
    // Copy data
    for(i = 0; i < n1; i++)
        L[i] = arr[low + i];
    for(j = 0; j < n2; j++)
        R[j] = arr[mid + 1 + j];
```

```

i = 0;
j = 0;
k = low;
// Merge arrays
while(i < n1 && j < n2) {
    if(L[i] <= R[j]) {
        arr[k] = L[i];
        i++;
    } else {
        arr[k] = R[j];
        j++;
    }
    k++;
}
// Remaining elements
while(i < n1) {
    arr[k] = L[i];
    i++; k++;
}
while(j < n2) {
    arr[k] = R[j];
    j++; k++;
}
}
void mergeSort(int arr[], int low, int high) {
    if(low < high) {
        int mid = (low + high) / 2;
        mergeSort(arr, low, mid);
        mergeSort(arr, mid + 1, high);
        merge(arr, low, mid, high);
    }
}
int main() {
    int arr[100], n, i;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    printf("Enter elements: ");
    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    mergeSort(arr, 0, n - 1);
    printf("Sorted array: ");
    for(i = 0; i < n; i++)
        printf("%d ", arr[i]);
    return 0;
}

```



Sample Input

```

5
38 27 43 3 9

```



Sample Output

```

3 9 27 38 43

```

Result

The given elements are successfully sorted using **Merge Sort**.

Conclusion

Merge Sort is an efficient sorting algorithm with **$O(n \log n)$** time complexity and is suitable for large datasets